

LESSON 3

Truth Tables

Name: _____ Date: _____

Why truth tables?

A **truth table** lists every way the parts of a compound statement can be true or false, and shows the result for the whole expression. With two basic statements p and q , there are **four** rows: TT, TF, FT, FF.

Recipe for building a table

1. Count variables: n variables $\rightarrow 2^n$ rows.
2. Write columns for each variable (often p changes slowest, q fastest — either pattern is fine if you are consistent).
3. Add columns for sub-expressions from the inside out (parentheses first).
4. The final column is the whole statement.

NOT

p	$\neg p$
T	F
F	T

AND

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

OR

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

WORKED EXAMPLE — $\neg(p \wedge q)$

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

English: “Not both” is false only when p and q are simultaneously true.

Tautology and contradiction (names worth knowing)

- **Tautology:** always true (final column all T). Example: $p \vee \neg p$.
- **Contradiction:** always false (final column all F). Example: $p \wedge \neg p$.

Three variables: With p, q, r you need 8 rows. Challenge problems may use three — work systematically so you do not skip a case.

Practice — Lesson 3

3.1 WARM-UP

Complete the missing cells in this OR table:

p	q	$p \vee q$
T	T	?
T	F	?
F	T	?
F	F	?

FILL T/F IN THE TABLE (OR REWRITE COMPLETED TABLE HERE)

3.2 WARM-UP

Build a full truth table for $\neg p \vee q$. How many rows are true in the final column?

TABLE + COUNT

3.3 STANDARD

Build a truth table for $(p \vee q) \wedge \neg p$. Describe in plain English when this compound is true.

TABLE + ENGLISH DESCRIPTION

3.4 STANDARD

Is $p \vee \neg p$ a tautology? Build the table and explain.

TABLE + YES/NO WITH REASON

3.5 STANDARD

Build truth tables for both sides and decide if they always match (always same final column):

$$\neg(p \wedge q) \quad \text{and} \quad (\neg p \vee \neg q)$$

TWO TABLES + CONCLUSION (THIS IS DE MORGAN'S LAW)

3.6 CHALLENGE

Build an 8-row truth table for $(p \wedge q) \vee (\neg p \wedge r)$. List every row where the final column is true.

FULL TABLE OR LIST OF TRUE ROWS

3.7

CHALLENGE

Design your own compound statement using $p, q, \neg, \wedge, \vee$ that is true in **exactly two** of the four rows. Write the expression and verify with a table.

EXPRESSION + VERIFICATION TABLE